

LOCAL LLM TEXT SUMMARIZATION USING STREAMLIT

Question 2 – Implementation

1. Question

Develop a Streamlit application for text summarization using a locally running Large Language Model.

2. Aim

To develop a Streamlit-based text summarization application that accepts user-provided text and generates a concise summary using a locally running Large Language Model through Ollama and Python.

3. Technologies Used

- Python 3.12.10
- Streamlit
- Ollama 0.33.2
- Llama 3.2 – local Large Language Model
- VS Code

4. Project Structure

Local_LLM_Text_Generation/

├── venv/ – Python virtual environment

├── app.py – Question 1 text generation application

└── summarization.py – Question 2 text summarization application

5. Implementation Steps

1. Use the existing Local_LLM_Text_Generation project folder.
2. Activate the existing virtual environment using `.\venv\Scripts\Activate.ps1`.
3. Reuse the installed Streamlit and Ollama Python packages.
4. Reuse the locally installed Llama 3.2 model through Ollama.
5. Create a new Python file named `summarization.py`.
6. Design the Streamlit interface with a model selector, text area, summary-length selector, and Generate Summary button.
7. Create a summarization prompt that instructs the local LLM to summarize only the supplied text.
8. Send the prompt to Llama 3.2 using the Ollama Python library.
9. Display the generated summary in the Streamlit interface.
10. Run the application using `streamlit run summarization.py`.

6. Python Implementation

```
import streamlit as st
import ollama
```

```
st.set_page_config(
    page_title="Local LLM Text Summarizer",
```

```

    page_icon="📄",
    layout="centered"
)

st.title("📄 Local LLM Text Summarizer")

st.write(
    "Summarize text using a locally running Large Language Model "
    "through Ollama."
)

model = st.selectbox(
    "Select Local LLM",
    ["llama3.2"]
)

text = st.text_area(
    "Enter text to summarize",
    placeholder="Paste the text you want to summarize here...",
    height=250
)

summary_length = st.selectbox(
    "Summary Length",
    ["Short", "Medium", "Detailed"]
)

if st.button("Generate Summary", type="primary", width="stretch"):

    if not text.strip():
        st.warning("Please enter some text to summarize.")

    else:
        if summary_length == "Short":
            instruction = "Summarize the text in 3 to 4 sentences."
        elif summary_length == "Medium":
            instruction = "Summarize the text in one clear paragraph."
        else:
            instruction = "Provide a detailed summary covering all important points."

        prompt = f"""
You are a professional text summarization assistant.

{instruction}

Do not add information that is not present in the original text.

Original Text:
{text}
"""

        response = ollama.chat(

```

```

        model=model,
        messages=[
            {"role": "user", "content": prompt}
        ]
    )

    summary = response["message"]["content"]

    st.subheader("Generated Summary")
    st.write(summary)

```

7. Working Principle

The user enters source text into the Streamlit text area and selects the required summary length. Python constructs a summarization prompt and sends it to the locally running Llama 3.2 model through Ollama. The model generates a summary based only on the supplied text, and Streamlit displays the result.

8. System Workflow

11. User enters the original text.
12. User selects the local LLM and summary length.
13. Streamlit captures the input.
14. Python creates the summarization instruction.
15. Ollama sends the request to Llama 3.2.
16. Llama 3.2 generates the summary.
17. The summary is returned to Python.
18. Streamlit displays the generated summary.

9. Input

Sample input: A paragraph explaining Artificial Intelligence, its applications, and the role of machine learning.

10. Output

The application generates a concise summary of the supplied text. In the demonstrated run, the Medium option produced a single clear paragraph covering the main ideas of Artificial Intelligence, its applications, and machine learning.

11. Application Output Screenshot

Summarize text using a locally running Large Language Model through Ollama.

Select Local LLM

llama3.2

Enter text to summarize

Artificial Intelligence is a branch of computer science that focuses on creating systems capable of performing tasks that normally require human intelligence. These tasks include learning, reasoning, problem solving, understanding natural language, recognizing images, and making decisions. AI is used in many areas such as healthcare, education, banking, transportation, and customer service. Machine learning is an important part of AI that enables computers to learn patterns from data and improve their performance over time.

Summary Length

Medium

Generate Summary

Generated Summary

Artificial Intelligence is a branch of computer science that creates systems capable of performing tasks that normally require human intelligence, such as learning, reasoning, and problem-solving. These systems are used in various areas, including healthcare, education, banking, transportation, and customer service, and rely on machine learning to learn patterns from data and improve their performance over time.

Figure 1: Streamlit interface showing local LLM text summarization

12. Result

Thus, a Streamlit application for text summarization was successfully developed using Python, Ollama, and the locally running Llama 3.2 Large Language Model. The application accepts source text, allows the user to choose a summary length, and displays the generated summary.

13. Conclusion

The implementation demonstrates how a local LLM can be used for automated text summarization without depending on a cloud-based language model API. The application provides a simple interface for generating short, medium, or detailed summaries from user-provided text.